# Exploratory Data Analysis and Visualization Report


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**WEEK 3 FINAL REPORT –CHURN ANALYSIS AND PRIDICTION MODELING**

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## 1. Overview of Week 3 Work

In Week 3, the main goal was to **predict student churn (drop-off)** using machine learning and to understand **why students disengage** from opportunities. This work builds on Week 1 (data cleaning and feature engineering) and Week 2 (exploratory data analysis).

Three machine learning models were built and compared:

- Logistic Regression
- Decision Tree
- Random Forest

The models were evaluated using standard performance metrics, and insights were generated to help improve student engagement and retention.

Whole Work Link:
https://drive.google.com/drive/folders/1LsKSj1Cri1Akvk27gliym8bA0Ws8Ocon?usp=sharing

Code File:
https://drive.google.com/file/d/12G_Te7BQRyDOc0XdZ0JBi-_HUj4MTbVa/view?usp=sharing

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## 2. Problem Statement

### Business Problem

Many students register for opportunities but do not complete them. This drop-off (churn) reduces program effectiveness and learner success.

### Prediction Problem

The problem is to **predict whether a student will churn or not**, using:

- Engagement behavior
- Demographic information
- Opportunity characteristics

This is a **binary classification problem**:

- Churn = 1 → student dropped / rejected / failed to start
- Churn = 0 → student completed or remained active

### 3. Dataset and What We Predict

#### Dataset Description

Each row in the dataset represents **one student's interaction with one opportunity**. The dataset includes:

- Student age, gender, country
- Engagement days and engagement score
- Opportunity category and duration
- Final status of the student

```
Dataset shape: (23, 28)

Column names:
[' Learner SignUp DateTime', 'Signup Month', 'Opportunity Id', 'Opportunity Name', 'Opportunity Category', 'First Name', 'Age',
'Date of Birth', 'Gender', 'Gender Binary', 'Country', 'Institution Name ', 'Institution Name', 'Current/Intended Major', 'Entr
y created at', 'Status Description', 'Status Code', 'Apply Date', 'Opportunity Start Date', 'Opportunity End Date', 'Engagment
Days', 'Opportunity_Duration_Level', 'Engagment Months', 'Churn', 'Engagement_Days_Normalized', 'Early_Drop', 'Engagement_Scor
e', 'Engagement_Level']
```

```
Missing values:
 Learner SignUp DateTime       0
Signup Month                   0
Opportunity Id                 0
Opportunity Name               0
Opportunity Category           0
First Name                     0
Age                            0
Date of Birth                  0
Gender                         0
Gender Binary                  0
Country                        0
Institution Name               0
Institution Name               0
Current/Intended Major         0
Entry created at               0
Status Description             0
Status Code                    0
Apply Date                     3
Opportunity Start Date        10
Opportunity End Date           0
Engagment Days                 0
Opportunity_Duration_Level     0
Engagment Months               0
Churn                          0
Engagement_Days_Normalized     0
Early_Drop                     0
Engagement_Score               0
Engagement_Level               0
dtype: int64
```

**What Is Being Predicted**

Using early engagement and background information, the models predict:

**Whether a student will churn before completing the opportunity**

**Where Problem Solving Happens in the Dataset**

- Engagement features show *how active a student is*
- Opportunity category shows *which programs have higher risk*
- Demographics help identify *at-risk groups*
- Churn label allows supervised learning

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**4. Data Preparation for Churn Analysis**

Before modeling, the dataset was cleaned and prepared:

- Missing values were handled
- Invalid and duplicate records were removed
- Date columns were validated
- Categorical variables were encoded
- Only pre-outcome features were used to avoid data leakage

**Engagement Score**

An **Engagement Score** was created using:

- Normalized engagement days
- Early-drop indicator

This score represents how actively a student participated before the outcome.

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**5. Machine Learning Models Used**

**1. Logistic Regression**

- Used as a baseline model
- Easy to interpret
- Shows how each feature increases or decreases churn risk

**2. Decision Tree**

- Captures non-linear relationships
- Easy to explain using rules
- Helps understand decision paths

**3. Random Forest**

- Combines multiple decision trees
- More robust and stable
- Handles complex feature interactions

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**6. Model Evaluation and Performance**

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

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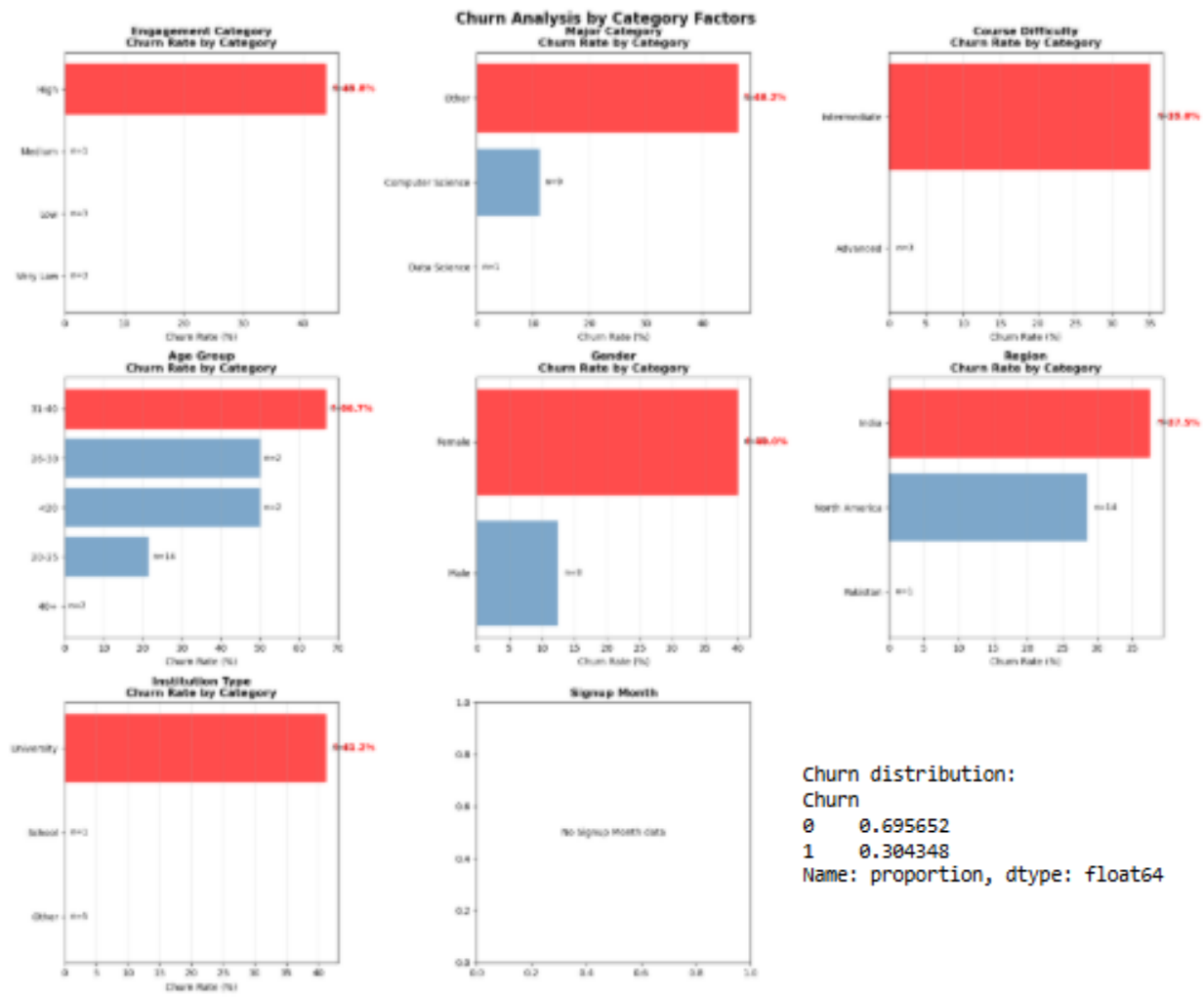
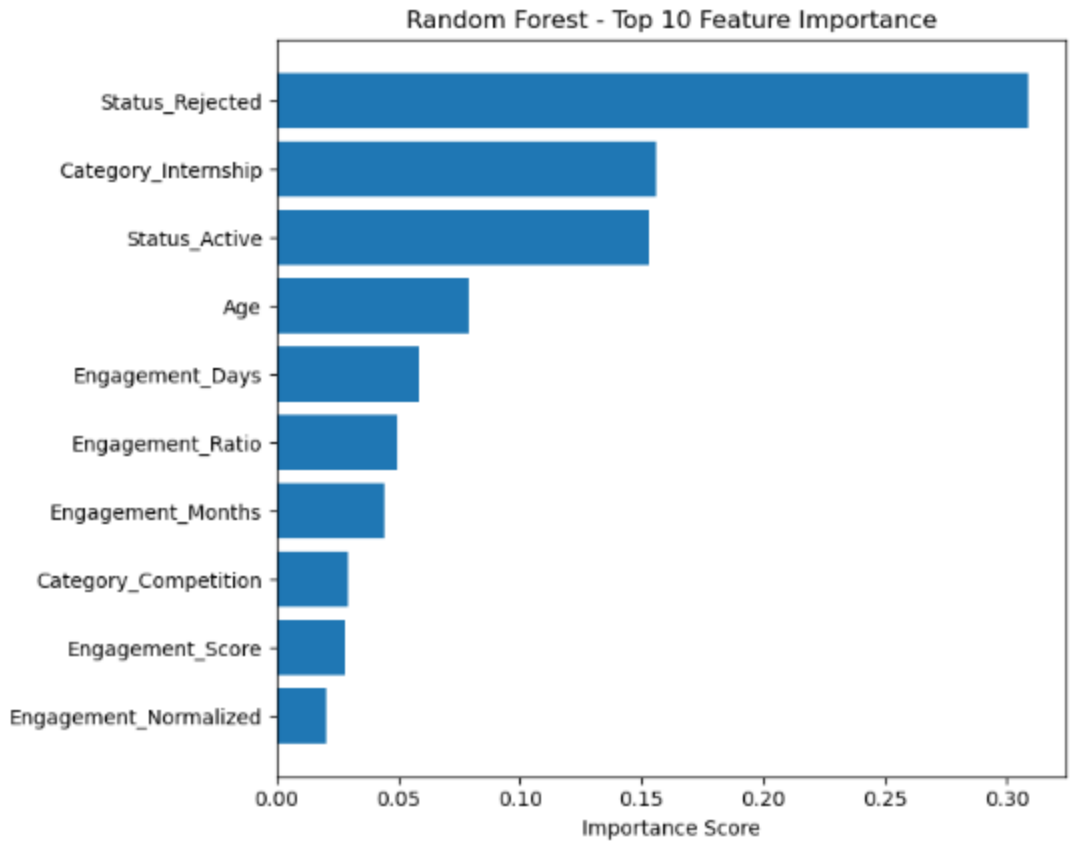
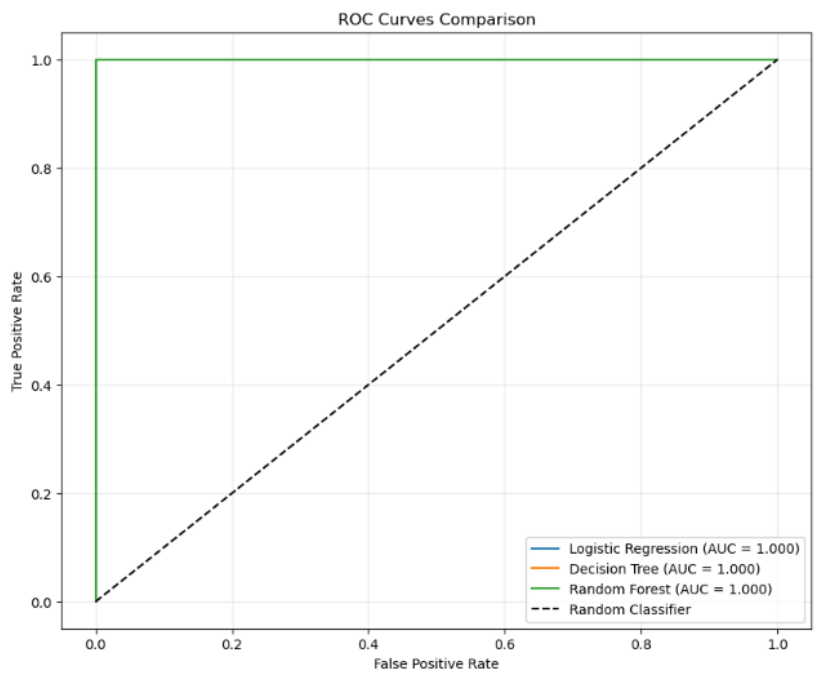
VISUALIZATION OF CATEGORY ANALYSIS
```




```
Churn distribution:
Churn
0    0.695652
1    0.304348
Name: proportion, dtype: float64
```


Random Forest - Top 10 Feature Importance
Status_Rejected
Category_Internship
Status_Active
Age
Engagement_Days
Engagement_Ratio
Engagement_Months
Category_Competition
Engagement_Score
Engagement_Normalized
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Importance Score


ROC Curves Comparison
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression (AUC = 1.000)
Decision Tree (AUC = 1.000)
Random Forest (AUC = 1.000)
Random Classifier

**Model Comparison (Summary)**

| Model | Performance | Notes |
|---|---|---|
| Logistic Regression | Very strong | Stable and interpretable |
| Decision Tree | Very strong | Easy to explain |
| Random Forest | Very strong | Robust but less interpretable |

Because the dataset is small, all models achieved very high scores. These results are treated as **proof of concept**, not final production performance.

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Model: Logistic Regression
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Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         3
           1       1.00      1.00      1.00         2

    accuracy                           1.00         5
   macro avg       1.00      1.00      1.00         5
weighted avg       1.00      1.00      1.00         5

ROC-AUC Score: 1.0000
Cross-validation ROC-AUC: 1.0000 (+/- 0.0000)
```

```
==================================================
Model: Decision Tree
==================================================

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         3
           1       1.00      1.00      1.00         2

    accuracy                           1.00         5
   macro avg       1.00      1.00      1.00         5
weighted avg       1.00      1.00      1.00         5

ROC-AUC Score: 1.0000
Cross-validation ROC-AUC: 0.9333 (+/- 0.2667)
```

```
==================================================
Model: Random Forest
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Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         3
           1       1.00      1.00      1.00         2

    accuracy                           1.00         5
   macro avg       1.00      1.00      1.00         5
weighted avg       1.00      1.00      1.00         5

ROC-AUC Score: 1.0000
Cross-validation ROC-AUC: 1.0000 (+/- 0.0000)
```

## 7. Why We Are Comfortable Moving Forward with the Selected Model

**Selected Model: Logistic Regression**

Reasons:

- High and stable performance
- Very easy to interpret
- Clear explanation of churn drivers
- Suitable for academic evaluation
- Easy to deploy and monitor

Although Random Forest also performed well, Logistic Regression provides the **best balance between accuracy and explainability**.

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================================================================================
MODEL COMPARISON
================================================================================

📊 MODEL PERFORMANCE COMPARISON (Sorted by Test F1-Score):
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| Model | Test Accuracy | Test Precision | Test Recall | Test F1-Score | Test ROC-AUC | Train Accuracy | Accuracy Diff |
|---|---|---|---|---|---|---|---|
| Logistic Regression | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0.000 |
| Decision Tree | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0.000 |
| Random Forest | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0.000 |

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🏆 BEST MODEL: Logistic Regression (F1-Score: 1.000)

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EVALUATING: Logistic Regression
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📊 PERFORMANCE METRICS:
Metric          Training     Test         Difference
-------------------------------------------------------
Accuracy        1.000        1.000        ≈0
Precision       1.000        1.000        ≈0
Recall          1.000        1.000        ≈0
F1              1.000        1.000        ≈0

📈 ROC-AUC Scores:
  Training: 1.000
  Test:     1.000
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**8. Interpretation of Results (Churn Analysis)**

**Key Drivers of Churn**

The most important factors linked to churn are:

- Low engagement score
- Early drop-off behavior
- Certain opportunity categories
- Specific age groups

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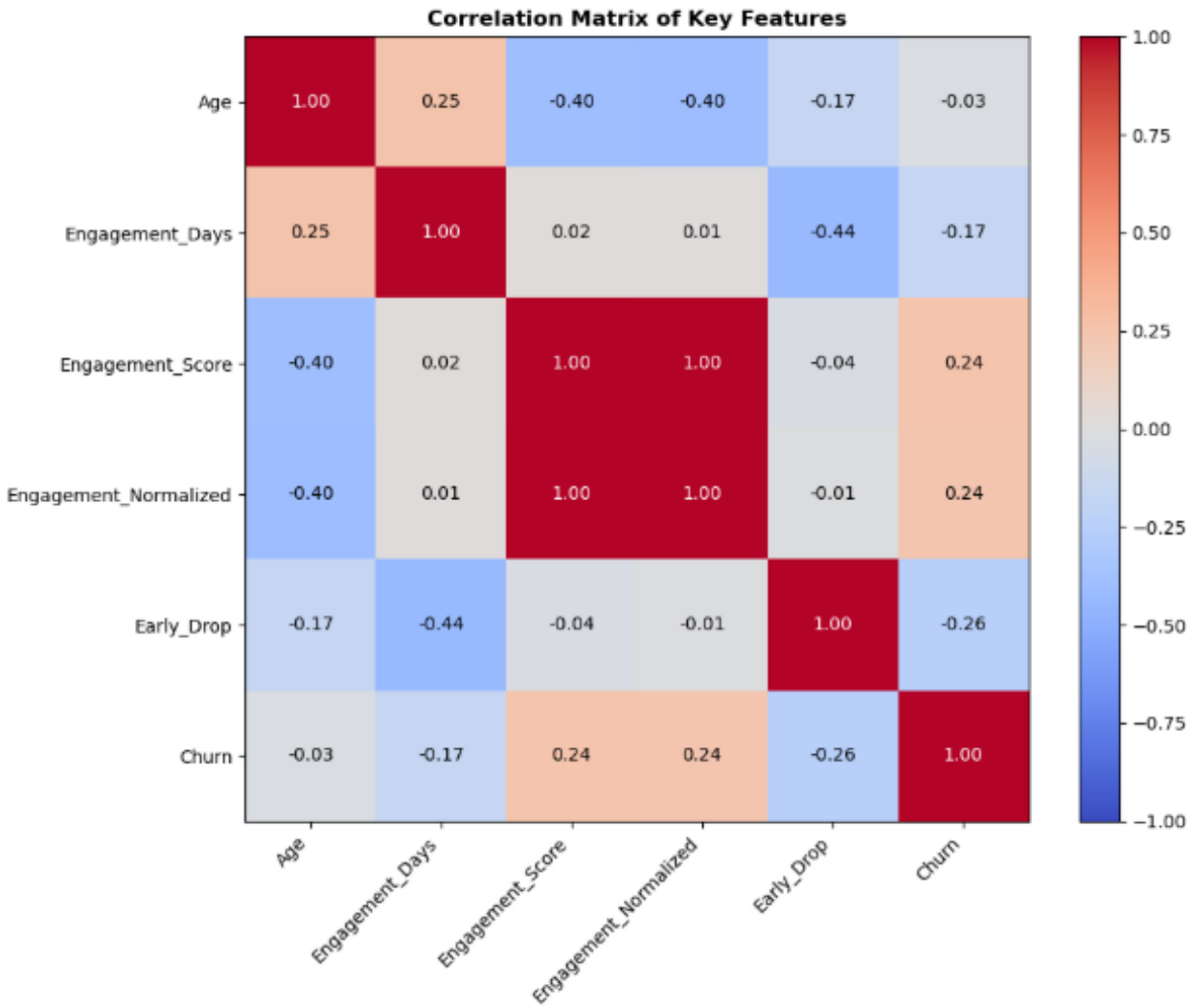
📈 CORRELATION ANALYSIS:
--------------------------------------------------
Correlation with Churn:
  Engagement_Score            0.244 (Weak Positive correlation)
  Engagement_Normalized       0.237 (Weak Positive correlation)
  Age                        -0.026 (Weak Negative correlation)
  Engagement_Days            -0.168 (Weak Negative correlation)
  Early_Drop                 -0.256 (Weak Negative correlation)
```

**What Problem Is Solved by Each Prediction**

- **Engagement prediction** → identifies students who need early support
- **Category prediction** → identifies high-risk programs
- **Demographic patterns** → helps target interventions




### 9. Visualization Explanation

The following visualizations were used:

- Model performance tables to compare accuracy and recall
- Feature importance charts to identify churn drivers
- Engagement vs churn charts to show behavior patterns

These visuals help explain results clearly to non-technical stakeholders.

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### 10. Actionable Insights

Key insights from the analysis:

- Students with very low early engagement are most likely to churn
- Medium-engagement students have the lowest churn risk
- Some opportunity categories consistently show higher churn

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### 11. Recommendations (Based on Past Experience)

#### Immediate Actions

- Introduce early-engagement alerts
- Provide extra support during the first few days

#### Process Improvements

- Track engagement daily
- Use churn scores to prioritize at-risk students

#### Long-Term Improvements

- Integrate churn prediction into the platform
- Continuously retrain models as more data becomes available
- Redesign high-churn opportunity categories

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🎯 PRIORITY ACTION PLAN:
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1️⃣ IMMEDIATE ACTIONS (Next 30 days):
   • Implement early dropout alerts for users with <30 days engagement
   • Create re-engagement campaign for rejected applicants
   • Set up monitoring for top 5 churn drivers

2️⃣ SHORT-TERM INITIATIVES (Next 90 days):
   • Develop engagement-boosting programs for high-risk segments
   • A/B test interventions for different major categories
   • Create personalized content based on demographic insights

3️⃣ STRATEGIC INITIATIVES (Next 6-12 months):
   • Redesign high-churn program categories
   • Implement predictive churn scoring system
   • Develop retention-focused curriculum changes

📊 MONITORING & MEASUREMENT:
   • Weekly: Track churn rate by top driver categories
   • Monthly: Review model performance and feature importance
   • Quarterly: Update analysis with new data and insights

💡 KEY INSIGHTS FOR STAKEHOLDERS:
   1. Engagement metrics are the strongest churn predictors
   2. Early interventions can significantly reduce churn
   3. Different segments require different retention strategies
   4. Continuous monitoring is essential for success
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**SUMMARY OF KEY FINDINGS**

**1. TOP CHURN DRIVERS IDENTIFIED:**

- Engagement metrics (days, score, early drop)
- Status indicators (rejection, dropout)
- Demographic factors (age)
- Program/category types

**2. CRITICAL INSIGHTS:**

- Early engagement is crucial for retention
- Different segments require tailored approaches
- Proactive intervention can prevent churn

**3. RECOMMENDED FOCUS AREAS:**

- Boost early-stage engagement
- Improve application experience
- Personalize based on demographics
- Monitor high-risk segments closely

**4. SUCCESS METRICS TO TRACK:**

- Reduction in early dropout rate
- Improved engagement scores
- Decreased churn in high-risk categories
- Increased retention across segments

**12. Limitations**

- Dataset size is small
- Perfect scores suggest possible overfitting
- Results should be treated as a **learning and demonstration exercise**

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## 13. Conclusion

This Week 3 analysis shows that machine learning can effectively predict student churn using engagement and demographic data. The project demonstrates the complete data science workflow, from data preparation to model evaluation and insight generation.

With more data and continued refinement, this approach can support scalable, data-driven strategies to improve student engagement and retention.